

Lecture 22

So far: Logic, Proof, Sets, Number Th, Graph Th, Counting.

Probability precise way to reason about uncertainty/risk/likelihood.

ex: the chance of a flush in poker is 2 in 1000.

the chance of a big earthquake in Berkeley by 2030 is 30%.

① what is chance?

② how do we calculate it?

Motivation: AI, stat, medicine, math, physics.

① experiment.

ex 1: Suppose I flip two fair coins (H or T with equal likelihood).
What is the chance the two are different?

All possible outcomes
successful outcomes

②

$$S = \{HH, HT, TH, TT\}$$
$$E = \{HT, TH\}$$

$$\therefore \text{probability of success} = \frac{\# \text{ of successful outcomes}}{\# \text{ of possible outcomes}} = \frac{2}{4} = \underline{\underline{1/2}}$$

Axioms of classical Probability

① An experiment is a well-defined procedure with a set of outcomes. An outcome is a complete specification of all of the relevant details of an expt.

② The set of possible outcomes is called the sample space S .

③ An event is a subset of the sample space

★ two further assumptions:

a) S is finite

b) All outcomes $s \in S$ are equally likely.

④ Given ★, the probability of E is

$$P(E) = \frac{|E|}{|S|}$$

What does $P(E)$ mean?

Frequentist Interpretation: If you repeat ① many times, the

outcome will lie in E a $P(E)$ fraction of the time.

ex 2:

Experiment: flip 10 fair coins.

equally likely.

Sample Space: $S = \{ (s_1, \dots, s_{10}) : s_i \in \{H, T\} \} = \{H, T\}^{10}$.

Event: $E = \{ (s_1, \dots, s_{10}) : \text{exactly } 5 s_i = H \}$.

Probability: $P(E) = |E|/|S| = \frac{\binom{10}{5}}{2^{10}} = \frac{252}{1000} \sim \underline{\underline{25\%}}$.

$$|S| = 2^{10}$$

$$|E| = \binom{10}{5}$$

ex 3: Recall : A deck has 13 cards from each of 4 suits: H, C, D, S.

A flush is a set of 5 ^{distinct.} cards from the same suit. ^{equal prob}

(Q) If I draw a hand of 5 cards uniformly at random from a deck, what is the probability of a flush?

modeling

Experiment: Draw an unordered hand of 5 cards.

Sample Space: $S = \{ T \subseteq \{1, \dots, 52\} : |T| = 5 \}$. $|S| = \binom{52}{5}$

Event:

$$E = E_H \cup E_C \cup E_D \cup E_S \quad \text{where}$$

$E_H = \{ T \in S : T \text{ contains only hearts} \}$ and

E_C, E_D, E_S are defined analogously.

$$|E| = 4 \binom{13}{5} \text{ by the sum rule.}$$

Probability: $P(E) = \frac{|E|}{|S|} = \frac{4 \binom{13}{5}}{\binom{52}{5}} = \frac{2}{1000}.$

Remark: Could have worked with $S = \{(t_1, \dots, t_5) : t_i \in [52], t_i \text{ distinct}\}$
but the order was irrelevant for E.

In general, choose S which:

- ① has all the relevant features
- ② makes counting techniques.

Ex4: What is the probability that 2 people in your section ⁽³⁰⁾_{people} have the same birthday?

(model of reality)

Experiment: Each person in the section is assigned a birthday $b_i \in \{1, \dots, 365\}$ uniformly at random.

Sample Space: $S = \{ (b_1, b_2, \dots, b_{30}) : b_i \in [365] \} = [365]^{30}$.
 $|S| = 365^{30}$

Event: $E = \{ (b_1, \dots, b_{30}) : \exists i \neq j \text{ s.t. } b_i = b_j \}$.

$\bar{E} = \{ (b_1, \dots, b_{30}) : b_i \text{ are distinct} \}$. $|\bar{E}| = P(365, 30)$
 $= 365! / 335!$

Probability: $P(E) = 1 - P(\bar{E}) \sim 1 - 0.1 = \underline{\underline{0.9}}$.

$$P(\bar{E}) = \frac{365 \cdot 364 \cdot \dots \cdot 336}{365 \cdot 365 \cdot \dots \cdot 365} \sim 0.1$$

$$\begin{aligned} \bar{E} &= S - E \\ P(E) + P(\bar{E}) &= \frac{|E|}{|S|} + \frac{|S - E|}{|S|} = 1. \end{aligned}$$

Remark: (Birthday Paradox) In any group of 23 people, there is a >50% chance that two have the same birthday.

(cf. 366 people needed to guarantee (100%) a coincidence by pigeonhole)

II Probability Theory

Motivation: What if outcomes are not equally likely?

Defn: A probability space consists of a sample space S together with a function $p: S \rightarrow \mathbb{R}$ such that

① $p(s) \in [0, 1]$ for all $s \in S$.

② $\sum_{s \in S} p(s) = 1$.

likelihood of s .

The function $p(s)$ is called a probability distribution.

An event is a subset $E \subseteq S$.

The probability of E is $P(E) = \sum_{s \in E} p(s)$

Remark: If all $s \in S$ are equally likely, this implies $p(s) = \frac{1}{|S|} \quad \forall s \in S$.
In this case, $P(E) = \sum_{s \in E} p(s) = \sum_{s \in E} \frac{1}{|S|} = |E|/|S|$.

ex 5:

Experiment: flip a biased coin, twice as likely to be H than T.

Sample Space: $S = \{H, T\}$.

Probability Dist:

$$p(H) = 2p(T)$$

$$p(H) + p(T) = 1 \rightarrow 3p(T) = 1$$

$$\rightarrow p(T) = 1/3, \\ p(H) = 2/3.$$

Monty Hall Problem

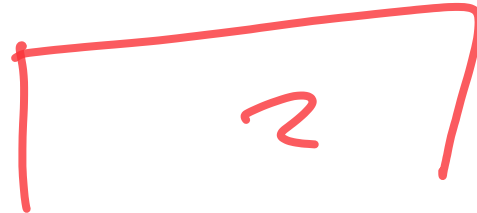
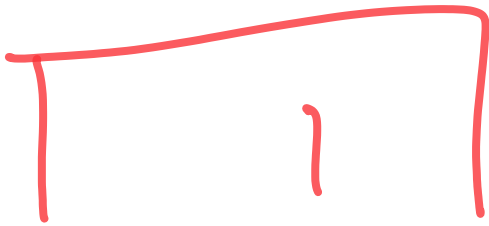
1) A **car** is placed v.a.r behind one door.
Goats are placed behind other doors.

2) You **choose** a door v.a.r.

3) Monty **opens** one of the other doors with a
goat v.a.r.

expt.

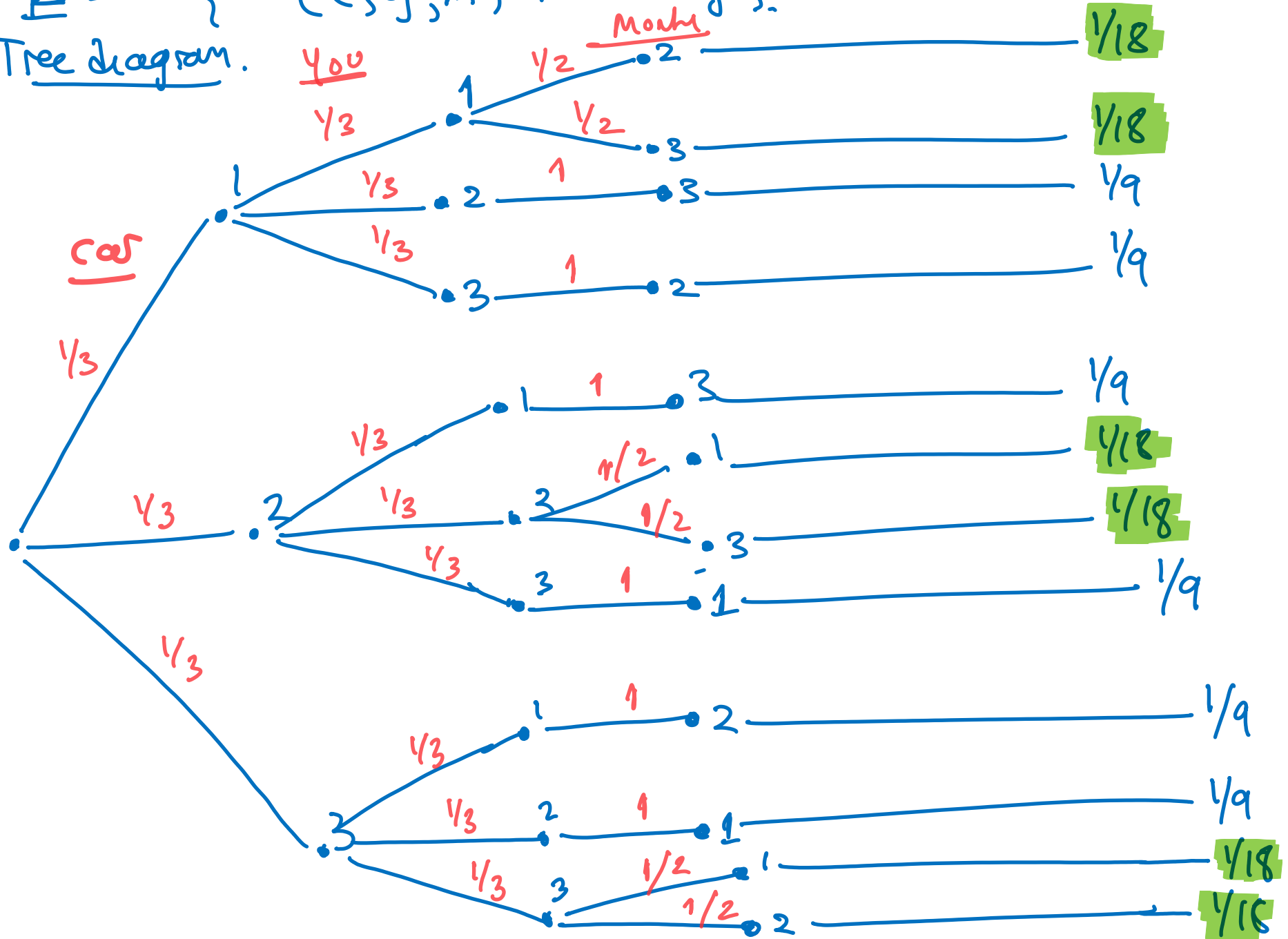
(Q) Should you switch?



$S = \{ (car, you, monty) : \begin{array}{l} monty \neq you, \\ monty \neq car, \end{array} \quad c, y, m \in \{1, 2, 3\} \}$

$E = \{ (c, y, m) : c = y \}$

Tree diagram.



$$P(E) = 6 \cdot \frac{1}{18} = \underline{\underline{\frac{1}{3}}}.$$

$$P(\bar{E}) = 1 - \frac{1}{3} = \underline{\underline{\frac{2}{3}}}. \quad \leftarrow \text{other door has the car.}$$

You should switch.

(Q) What about 1000 doors?

$$P(E) = \frac{1}{1000}. \quad \text{door \#1, Monty opens \#1000.}$$

$$P(\bar{E}) = \frac{999}{1000} = P(\text{Car is behind } 2 \vee 3 \vee 4 \dots \vee 999)$$

$$P(\text{door 2}) = \frac{999}{1000} \cdot \frac{1}{998} > \frac{1}{1000}.$$

So far: One event, calculate $P(E)$.

Next theorems: Given $E_1, E_2, \dots, E_k \subseteq S$,
given $P(E_1), \dots, P(E_k)$
what can we deduce about $P(F)$?

Complement Rule (Above).

Sum Rule: If E, F are events then

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

Proof:
$$P(E \cup F) = \sum_{S \in E \cup F} p(S) = \sum_{S \in E} p(S) + \sum_{S \in F} p(S) - \sum_{S \in E \cap F} p(S). \quad \square$$

Remark: If $E \cap F = \emptyset$ then $P(E \cup F) = P(E) + P(F)$.
 $\uparrow$ mutually exclusive.